INCH-POUND

MIL-DTL-17/208A
12 June 2014
SUPERSEDING
MIL-C-17/208
29 September 1993

DETAIL SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 185 OHMS,
M17/208-00001

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall
consist of this specification sheet and MIL-DTL-17.

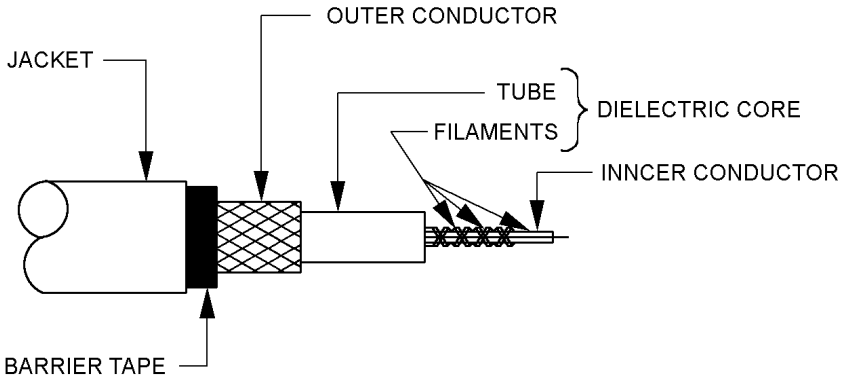


FIGURE 1. General configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Solid silver coated copper wire. Overall diameter: .007 inch ± .0010.
Dielectric core	Type A-3: Air-spaced polyethylene. A braid of 24 polyethylene filament threads, .020 inch approximate diameter each, with 2 picks/inch approximate, under an extruded tube. Overall diameter: .285 inch ± .010. Alternate Type A-3: Air-spaced polyethylene: A braid of 8 polyethylene filament threads, .035 inch approximate diameter each, with 2 picks/inch approximate, under an extruded tube. Overall diameter: .285 inch ± .010.
Outer conductor	Single braid of AWG#34, bare, copper wire. Diameter: 340 inch ± .340. Coverage: 91.1% Carriers: 24 Ends: 8 Picks/inch: 7.0 ± 10%

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TABLE I. Description – Continued.

Components	Construction details
Barrier tape	A .001 inch thick polyester tape faced with a .002 inch thick layer of aluminum. The tape will be applied with a 50% lap, minimum. Aluminum face toward the outer conductor. Diameter: .350 inch, maximum.
Jacket	Cross-linked polyolefin. Diameter: .405 inch ± .010.

REQUIREMENTS:

Continuous working voltage: 1,000 V rms, maximum.

Operating frequency: 1 GHz, maximum.

Velocity of propagation: 88 percent, nominal.

Power ratings: See figure 2.

Operating temperature range: -40°C to +80°C.

Inner conductor properties:

DC resistance (maximum at +20°C): 54.4 ohms per 100 feet, maximum.

Elongation: 2 percent, minimum.

Tensile strength: 110 klbf/inch/inch, minimum.

Engineering notes: This cable is useful in low capacitance, medium low temperature applications. (See connector series “N”, “C”, and “SC” per MIL-C-39012). Use this cable in new designs in –lieu-of MIL-C-17/47 cables.

The US Government preferred system of measurement is the metric SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound, the inch-pound units shall take predecence.

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination: Applicable.

Out-of-roundness: Not applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors: Not applicable.

Aging stability: +98° ± 2°C.

Cold bend: -40°C ± 2°C.

Stress crack resistance: Not applicable.

Dimensional stability: Not applicable.

Contamination: Not applicable.

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Flame propagation: Applicable.

Acid gas generation: 2.0 percent, maximum.

Halogen content: 0.2 percent, maximum.

Immersion test:

Tensile strength, percent of unaged minimum: 50.

Elongation, percent of unaged minimum: 50.

Smoke index: 25 maximum.

Toxicity index: 5 maximum.

Durometer hardness: (Type A) 80 minimum.

Weathering: Applicable.

Abrasion resistance: 75 cycles, minimum (jacket only).

Tear strength: 35 pounds per inch minimum.

Heat distortion: 30 percent maximum distortion.

Physical tests on unaged jacket:

Tensile strength: 1,300 psi, minimum.

Elongation: 160 percent minimum.

Physical tests on aged jacket:

Air oven:

Tensile strength, percent minimum: 60.

Elongatin, percent minimum: 60.

Hot oil immersion:

Tensile strength, percent minimum: 50.

Elongation, percent minimum: 50.

Tensile strength and elongation: 1,300 psi, 160 percent minimum.

Weight: 8.9 pounds per 100 feet maximum.

Electrical:

Spark test: 5,000 V rms, minimum.

Voltage withstanding: 5,000 V rms, minimum.

Insulation resistance: Not applicable.

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Corona extinction voltage: Not applicable.

Characteristic impedance: 185 ± 10.

Attenuation: 8.5 dB per 100 feet, maximum at .4 GHz.

Structural return loss: Not applicable.

Capacitance: 7.2 pF per foot, maximum.

Capacitance stability: ± 5 percent.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise: Not applicable.

Time delay: Not applicable.

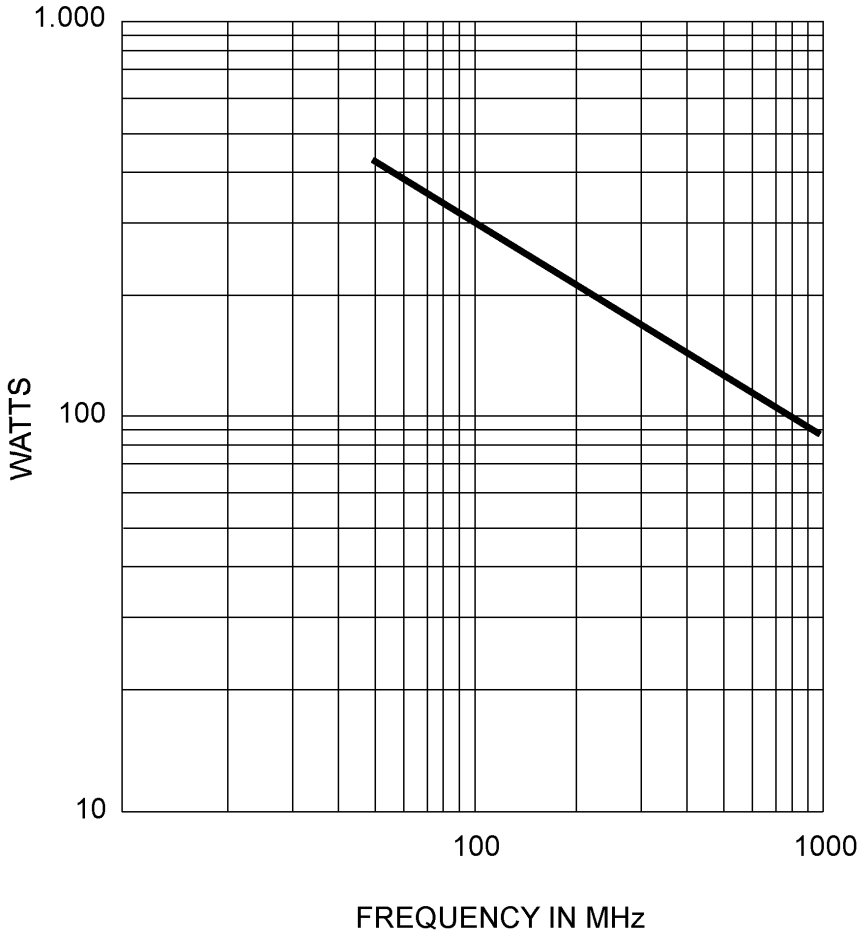


FIGURE 2. Power rating at 25°C sea level.

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Part or Identifying Number (PIN): M17/208-00001.

Referenced documents. This document references MIL-DTL-17.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA-CC

Review activities:
Army – AR, AT, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19, 99

(Project 6145-2014-011)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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